Mobile Sales Data Analysis Using Power BI

1. Executive Summary

This project aims to develop an interactive dashboard using Power BI to analyze mobile sales data and customer purchasing behavior. The analysis focuses on mobile brands, units sold, sales revenue, customer ratings, payment methods, customer demographics, cities, and mobile models. The dashboard helps businesses understand sales performance and customer preferences to support better decision-making. By integrating multiple datasets and a calendar table, the project enables advanced business intelligence and time-based analysis.

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2. Problem Statement

Background

Mobile retail businesses collect a large amount of transaction data every day. Without proper analysis, it becomes difficult to identify top-selling brands, understand customer behavior, evaluate sales trends, and measure customer satisfaction.

Objective

The objective of this project is to analyze mobile sales data using Power BI and generate meaningful insights regarding sales performance, customer behavior, brand popularity, payment trends, and customer demographics.

Scope

- Brand-wise Sales Analysis
- Mobile Model Performance Analysis
- Units Sold and Revenue Analysis
- Customer Age Group Analysis
- Customer Purchase Behavior Analysis
- City-wise Sales Analysis
- Payment Method Analysis
- Customer Rating Analysis
- Monthly and Yearly Sales Trends
- Time Intelligence Analysis
- KPI Monitoring Dashboard
- Multi-Sheet Data Analysis

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3. Data Sources

Primary Dataset (Sheet 1: Mobile Sales Data)

- Transaction ID
- Day
- Month
- Year
- Day Name
- Brand
- Units Sold
- Price Per Unit
- City
- Payment Method
- Customer Ratings
- Mobile Model

Secondary Dataset (Sheet 2: Customer Information)

- Transaction ID
- Customer Name
- Customer Age

Additional Dataset (Sheet 3)

- Additional business-related information used for advanced reporting, analysis, and dashboard enhancement.

Calendar Table

The Calendar Table is created for advanced time-based analysis and contains:

- Date
- Day
- Month
- Month Name
- Quarter
- Year
- Week Number

Data Relationships

- Transaction ID is used to connect the datasets.
- Calendar Table is connected with transaction dates for time intelligence calculations.

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4. Methodology

Project Feasibility

The project is feasible because the dataset contains complete mobile transaction details, customer information, and sales-related attributes required for analysis.

Data Cleaning and Preprocessing

1. Handling Missing Values
2. Removing Duplicate Records based on Transaction ID
3. Data Type Conversion for Numeric and Date Fields
4. Standardizing Brand, City, Payment Method, and Mobile Model Formats
5. Date Formatting for Monthly and Yearly Trend Analysis
6. Outlier Detection in Units Sold, Revenue, and Customer Ratings
7. Creating Calculated Columns such as Total Revenue and Age Groups
8. Data Validation to Ensure Data Accuracy

Data Modeling

1. Creating Relationships Between Multiple Datasets
2. Building Calendar Table
3. Creating DAX Measures and KPIs
4. Optimizing Data Model Performance

Tools Used for Data Cleaning

- Microsoft Excel
- Power Query in Power BI
- SQL Queries

Dashboard Design

- KPI Cards (Total Sales, Revenue, Units Sold, Average Rating)
- Brand Performance Analysis
- Mobile Model Analysis
- Customer Age Group Analysis
- Payment Method Analysis
- Monthly Sales Trend Analysis
- Yearly Sales Trend Analysis
- City-wise Sales Analysis using Maps
- Interactive Filters and Slicers

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5. Expected Outcomes

The project will provide an interactive dashboard to understand mobile sales performance and customer purchasing behavior.

Business Questions

- Which mobile brands generate the highest revenue?
- Which mobile models are sold the most?
- Which cities generate maximum sales?
- Which customer age group purchases more mobiles?
- How do customer ratings affect sales performance?
- Which payment method is preferred by customers?
- What are the monthly and yearly sales trends?
- Which brands perform better in the market?
- What is the quarterly sales performance?
- What is the year-over-year sales growth?
- How does customer age influence purchasing behavior?

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6. Tools and Technologies

- Power BI – Dashboard Development and Visualization
- Microsoft Excel – Data Cleaning and Preparation
- SQL – Data Analysis and Querying
- Power Query – Data Transformation
- DAX – KPI Creation and Calculations
- Calendar Table – Time Intelligence Analysis

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7. Challenges and Future Enhancements

Challenges

- Handling Missing or Inconsistent Data
- Managing Multiple Datasets Efficiently
- Creating Complex DAX Calculations
- Maintaining Dashboard Performance
- Ensuring Accurate Data Relationships

Future Enhancements

- Sales Forecasting using Historical Data
- Customer Purchase Prediction
- Customer Segmentation Analysis
- Advanced Customer Satisfaction Analysis

• AI-Based Product Recommendation System
• Online Dashboard Publishing in Power BI Service
• Real-Time Dashboard Integration

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8. Conclusion

This project aims to provide a detailed Mobile Sales Data Analysis Dashboard using Power BI. The dashboard will help analyze sales performance, customer behavior, city-wise trends, payment preferences, and brand performance. The integration of multiple datasets and a calendar table enables advanced reporting and time intelligence analysis. The project supports data-driven decision-making through meaningful visualizations and business intelligence techniques.

Project Information

Project Title: Mobile Sales Data Analysis Using Power BI

Domain: Data Analytics / Business Analytics

Tools Used: Power BI, Excel, SQL, Power Query, DAX

Dataset Used: Mobile Sales Dataset, Customer Information Dataset, Additional Business Dataset, and Calendar Table